S. Parisa Dajkhosh

PhD Candidate, Electrical & Computer Engineering

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Summary

PhD candidate in Electrical & Computer Engineering at the **University of Memphis** (expected 2027). PhD research focuses on UAV perception, navigation, and geo-localization for stream assessment and environmental monitoring, including multimodal aerial perception and field applications. M.Sc. focused on physics-informed deep learning, image reconstruction and denoising, unrolled optimization, and generative models for inverse problems in the Computational Imaging Research Lab (CIRL, 2021 - 2023). Published and presented at **MDPI Remote Sensing**, **SPIE**, and **OPTICA**. Open to **industry R&D, research scientist, and postdoc** roles starting late 2027.

- **Experience:** 5 yrs graduate research (UoM CIRL & RSL) · 1 yr industry R&D (Part AI) · 8+ yrs teaching programming & robotics · robotics competitions & awards since 2010.
- **Recognition:** Herff Fellowship, fully funded 5 consecutive semesters (2022 - 25) · 2nd place, SICK LiDAR Challenge, USA & Canada (2026) · Grant for autonomous UAV transmission-line inspection (2017) · Top 0.2% (rank 609), national entrance exam.
- **Stack:** Python (Ultralytics, PyTorch, TensorFlow, OpenCV, etc.), MATLAB/Simulink, C/C++, Verilog, FPGA, embedded systems, Linux, Git, LaTeX, nixOS, electrical circuit design.

Education

Ph.D., Electrical & Computer Engineering, *University of Memphis*, Memphis, TN *Aug 2023 - 2027 (expected)*

Concentration: Computer Engineering
Fully funded through graduate assistantship. GPA: **3.98/4.00**. Advisor: Dr. E. Jacobs.
Dissertation: *TerrAInav: Vision-AI-Based UAV Navigation, SLAM, and Geo-Localization for Environmental Monitoring.*
Selected courses: Linear Systems Analysis, Bayesian Inference, Fundamentals of Data Science.

M.Sc., Electrical & Computer Engineering, *University of Memphis*, Memphis, TN *Aug 2021 - Jul 2023*

Concentration: Computer Engineering
Fully funded. GPA: **3.96/4.00**. Advisor: Dr. C. Preza.
Thesis: *Image Reconstruction for Structured Illumination Microscopy using Physics-Guided DNNs (UPIGAN, Unrolled Physics-Informed GAN).*
Selected courses: Neural Networks, Random Signals and Noise, Image Processing, Data Mining, Multimedia Systems, Artificial Intelligence, Advanced Machine Learning.

B.Sc., Electrical Engineering (Concentration in Control Engineering), *Amirkabir University of Technology (Tehran Polytechnic)*, Tehran *Sep 2013 - Feb 2018*

Honors admission - one of the top-10 entering students in the program.
Thesis: *Indoor Autonomous Landing of a Quadrotor with Video Processing.* Advisor: Dr. F. Abdollahi.
Selected courses: Linear Algebra, Advanced Programming, Computational Intelligence, Control of Linear Systems, Digital Control, Modern Control, Signals & Systems, Computer Architecture.

Skills

- **Programming:** Python (Ultralytics, PyTorch, Torch-Vision, TensorFlow, NumPy, Pandas, SciPy, Scikit-learn, OpenCV, Keras), MATLAB/Simulink, C, C++, Verilog, Bash, Arduino, Qt.
- **Deep learning:** GANs, physics-informed networks, unrolled optimization, attention mechanisms, super-resolution, image-to-image translation, multimodal models, promptable and open-vocabulary models (text/box-prompted detection and segmentation), detector-segmenter pipelines, prompt-ablation studies.
- **Edge AI & benchmarking:** NVIDIA Jetson Orin NX deployment, accuracy-latency profiling, reproducible configuration-driven evaluation.
- **Computer vision & imaging:** feature extraction, object detection, OpenCV, ARUCO, optical-transfer-function modeling, structured illumination microscopy, camera calibration, geo-localization.
- **Hardware & embedded:** Raspberry Pi, NVIDIA Jetson, Khadas, Pixhawk, FPGA (Xilinx ISE), CPLDs, ARM Cortex-M4 (STM32, Keil), AVR, KiCad, Altium Designer.
- **Tools & DevOps:** Git, nixOS, Docker, Linux, LaTeX, LabVIEW, Scrum/Kanban.

Journal Publications

S. P. Dajkhosh, O. Furxhi, P. M. Le, E. L. Jacobs. *"TerrAInav Sim: An Open-Source Simulation of UAV Aerial Imaging from Satellite Data."* **Remote Sensing, MDPI**, 2025.

A. Gimeno-Gomez, **S. P. Dajkhosh**, M. S. H. Younis, C. T. S. Van, C. Preza, J. C. Barreiro-Hervas, G. Saavedra. *"Management of the axial modulation of the illumination pattern in structured illumination microscopy using an extended illumination source."* **Optics Express**, 2023.

Proceeding Publications and Technical Reports

S. P. Dajkhosh (presenter), P. M. Le, L. D. Poloni, J. A. Tuffaha, E. L. Jacobs. *"Vision-Based Uncrewed Aerial System for Real-Time Stream Assessment and Environmental Monitoring."* **SPIE Defense + Security, MD**, 2026.

S. P. Dajkhosh (presenter), D. G. Ortega, M. Davoodi, E. L. Jacobs. *"UAV Navigation for Stream Data Collection Through Tree Canopies Using Dynamic Motion Primitives."* **SPIE Defense + Commercial Sensing, FL**, 2025.

S. P. Dajkhosh (presenter), O. Furxhi, C. K. Renshaw, E. L. Jacobs. *"Aerial Image Feature Mapping using Deep Neural Networks."* **SPIE Defense + Commercial Sensing, MD**, 2024.

S. P. Dajkhosh (presenter), M. Hossain, C. Preza. *"USR: Unrolled Super-Resolution with Deep Priors for Structured Illumination Microscopy."* **OPTICA / Frontiers in Optics, MA**, 2023.

S. P. Dajkhosh (presenter), M. Hossain, C. Preza. *"Image Reconstruction for Structured Illumination Microscopy using an Unrolled GAN."* **SPIE Defense + Commercial Sensing, FL**, 2023.

MM. Rahimi, MM. Shirazi, S. P. Dajkhosh, *et al.* *"Parsian Extended Team Description."* **Team Description Paper, RoboCup 2016, Leipzig, Germany. SSL Senior League. (Peer-reviewed)**.

Manuscripts in Preparation

S. P. Dajkhosh, L. D. Poloni, X. Huang, E. L. Jacobs. *"Anait-Bench: An Open-Source Framework to Benchmark Conditioned Perception, with a Case Study in UAV Stream Monitoring."* **In preparation; target: IEEE Transactions on Geoscience and Remote Sensing (TGRS)**.

S. P. Dajkhosh, L. D. Poloni, X. Huang, E. L. Jacobs. *"Anait-Eye: A Joint Real-Time Model for Water-Mask Segmentation and Litter Detection with a UAV Stream-Corridor Dataset."* **In preparation;**

target: IEEE Transactions on Geoscience and Remote Sensing (TGRS).

S. P. Dajkhosh, L. D. Poloni, M. Davoodi, X. Huang, E. L. Jacobs. *"StreamEye: UAV-Based Real-Time Environmental Stream Assessment."* **In preparation**.

S. P. Dajkhosh, O. Furxhi, C. K. Renshaw, E. Jacobs. *"TerrAInav Mapper: Aerial Imaging Feature Extraction Using Siamese Networks."* **In preparation**.

Research Experience

Graduate Assistant, Doctoral Researcher, *Remote Sensing Lab (RSL), University of Memphis*, Memphis, TN — *Aug 2023 - present*

- **TerrAInav (Vision-Based Navigation, and Geolocalization):** multimodal field-applied research on UAVs including path planning with **Dynamic Movement Primitives (DMPs)**, SLAM, real-time water-body mapping, and multi-sensor aerial data collection; deep-learning image registration of UAV-captured aerial imagery; and geo-localization for visual stream assessment and environmental monitoring.
- **Anait-Bench:** open-source framework that benchmarks promptable and non-promptable detectors and segmenters, alone and as detector-segmenter pipelines, under text, box, and no-prompt conditions with uninformative-prompt controls; measures accuracy and latency on a workstation and verifies the most promising configurations on an NVIDIA Jetson Orin NX.
- **Stack:** Python (Ultralytics, PyTorch, Torch-Vision, TensorFlow, OpenCV, etc.), MATLAB, LaTeX, nixOS, embedded systems, electrical circuit design.

Graduate Assistant, Master's Researcher, *Computational Imaging Research Lab (CIRL), University of Memphis*, Memphis, TN — *Aug 2021 - Jul 2023*

- Built **UPIGAN**, an unrolled, physics-informed GAN that bakes the SIM forward model and OTF into the network for 3D super-resolution microscopy. Outperformed baseline channel-attention GANs on real SIM datasets. Led a sub-team of three in collaboration with Dr. C. Van.
- Using MATLAB, simulated new enhancements in the optical transfer functions for wide-field, tunable, and 3-Wave SIM microscope geometries, in a team of 5 with the **University of Valencia**
- **Stack:** TensorFlow, OpenCV, physics guided deep learning, 3D image super resolution, MATLAB

Volunteer Research Collaborator, *University of New South Wales*, Remote — *Aug 2020 - Jun 2021*

- Statistical analysis of a genomics dataset under Dr. Alinajad-Rokni: hypergeometric, chi-square, Fisher exact, and z-score specificity tests; hierarchical clustering in Python to identify tissue-specific and disease-causative genes.
- **Stack:** Python (NumPy, Pandas, Scikit-learn), MATLAB, Clustering Algorithms.

Volunteer Research Collaborator, *Institute for Cognitive & Brain Sciences, Shahid Beheshti University*, Tehran — *Feb 2018 - Aug 2019*

- Built a Python-based human - computer interface for **Cyberball** and **Iowa Gambling Task** experiments under Dr. Borhani, studying the effect of social rejection on risk-taking.

Undergraduate Research Assistant, *Control of Multi-Vehicle Systems (CMVS) Lab, Amirkabir University of Technology*, Tehran — *Jan 2016 - Aug 2019*

- Thesis project (A+): autonomous indoor landing of a quad-copter on a marker pad using video processing on a Raspberry Pi (OpenCV + ARUCO).
- Collaborated in a team of ∼6 for full implementation of a custom VTOL drone design.

- Designed independent PID controllers for each quad-copter degree of freedom; bridged C++ vision code to MATLAB/Simulink for control synthesis.

Undergraduate Research Assistant, *Parsian Robotics Lab, Amirkabir University of Technology*, Tehran *Jan 2013 - Jan 2016*

- **Small Size Soccer League:** Led electronics team of 3-4. Collaborated in a team of 10-15. Designed a 16.8 V to 200 V boost DC-DC converter, two-way NRF wireless link, and implemented a soft-core microprocessor on FPGA in Verilog. Full implementation of a FPGA/ARM processor with motor controllers for MIROSOT Soccer League.
- **Quad-copter internship:** Designed a fully integrated autonomous flight board (processor, motor drivers, IMU, NRF link, vision). Advisor: Dr. M. Khosravi.

Industry Experience

Lead Robotics & API Developer, *Part AI Research Center*, Tehran *Sep 2020 - Aug 2021*

- **Robotics:** designed and built cooperative agricultural quad-copters from scratch, leading a team of 4. Collaborated with the project manager on task definition, purchasing decisions, and team priorities.
- **Embedded systems:** implemented **YOLO** and **YOLO-Lite** object-detection on Khadas and NVIDIA Jetson dev kits; led cross-team consultations for embedded-systems integration.
- **AI software:** authored Python AI APIs for **Microsoft AirSim** flight simulator; designed the software architecture and class diagrams for the simulation API surface.
- **Process:** ran the team's Kanban + Scrum board on GitLab.
- **Stack:** Sensor fusion, Python APIs, MATLAB, Git, Arduino, embedded systems, Docker, IoT, robotics, Linux, Scrum, Kanban.

Teaching & Mentoring

Instructor of Record, EECE 2207: Engineering Mathematics Applications, *University of Memphis*, Memphis, TN *Sep 2024 - May 2025*

Taught Introduction to MATLAB and its applications in engineering math.

Mentor, Vertically Integrated Projects, *University of Memphis*, Memphis, TN *Sep 2022 - present*

Vertically Integrated Projects (VIP) Mentor (continual): mentoring undergraduates individually and in teams of 4-6; research methodology, paper reading & writing, programming in Python and MATLAB, electronics circuit and 3D mechanical design and fabrication, etc. In collaboration with Dr. S. Ivey and Dr. E. Jacobs.

Teaching Assistant, *Neuromatch Academy*, (Remote) *Jul - Aug 2021*

TA'd both the **Computational Neuroscience** and **Deep Learning** summer schools, with hands-on Python coding (NumPy, SciPy, PyTorch) for international cohorts.

Programming & Robotics Teacher, *Farzanegan 1 Middle School (NODET)*, Tehran *Jul 2013 - May 2021*

Taught Python, C/C++, Qt, Small Basic, and Scratch; led a graph-theory workshop; mentored teams to national & international competition wins. Taught analog/digital circuit design and AVR robotics from scratch.

Workshop Instructor, MATLAB & Simulink, *Al Zahra University*, Tehran *Jul - Aug 2020*

Workshop Instructor, Python, *Sharif University of Technology Business Summer School*, *Jun 2020* Tehran

Workshop Instructor, FPGA & Verilog, *Amirkabir University of Technology*, Tehran *2018 - 2019*

Lab Instructor & Teaching Assistant, Computational Intelligence, *Amirkabir University of Technology*, Tehran *Jan 2018 - Aug 2019*

Co-designed the lab plan; supervised modeling and control of a UGV using fuzzy logic and neural networks; graded assignments and reports.

Mentor, Quera Junior C++ Workshop (online, ages 12 - 16), *Quera* *Jun - Aug 2020*

Honors, Awards & Grants

- **Graduate Herff Fellowship**, Herff College of Engineering, University of Memphis - 5 consecutive semesters, 2022 - 2025. Coordinator: Dr. R. Meier.
- **2nd Place, 8th Annual EECE Department Poster Competition**, University of Memphis, 2022.
- **2nd Place**, SICK LiDAR StreamEye: LiDAR Challenge across USA and Canada, Proposing Multi-UAS Visual Stream Assessment, 2025 - 26.
- **Grant Recipient**, National Science & Technology Research Grant. Project: Autonomous Quadcopter with Auto-charge for Inspection of Transmission Lines. Amirkabir University, 2017.
- **Participant**, International RoboCup Senior Soccer Small Size League (one of ∼20 elite international university teams; contributed to circuit design and FPGA programming). Hefei / João Pessoa / Leipzig, 2014 - 2016.
- **Participant**, International RoboCup Regional Open Senior Soccer Small Size League, 2015 - 16.
- **Honors Admission**, one of the top-10 entering students in the Electrical Engineering program at Amirkabir University, 2013.
- **Ranked 609 (top 0.2%)**, Nationwide university entrance exam, 2013.
- **3rd Place**, International RoboCup Regional Open Junior Soccer 2×2 Lightweight Robots, 2012.
- **4th Place**, International RoboCup Junior Soccer 2×2 Robots, Istanbul, Turkey, 2011 (Individual & Super-team leagues; oral and poster presentation).
- **1st Place**, Khwarizmi Competition Junior Soccer Robots, Amirkabir University, 2010.

Certifications & Professional Development

- **FAA Remote Pilot Certificate (Part 107)**, effective June 2026.
- **Graduate Certificate in Data Science**, University of Memphis, 2021 - 2024.
- **Referee**, NODET Young Researchers' Festival, Engineering & Technology section, Tehran, 2018.
- **Organization Committee & Referee**, FarzCup Robotic Competition, 2012 - 2014.
- **Computational Neuroscience Summer School**, Neuromatch Academy, 2020.
- **Reinforcement Learning Winter School**, Loop Academy, 2019.

Talks & Posters

- *Contributor*, A. Gimeno-Gomez, **S. P. Dajkhosh**, M. S. H. Younis, C. T. S. Van, C. Preza, J. C. Barreiro-Hervas, G. Saavedra. *"Performance Comparison of Tunable Wollaston-Based and Three-Wave Interference Structured Illumination Microscopes."* **Focus on Microscopy**, 2023.
- *Contributor*, A. Gimeno-Gomez, C. T. S. Van, **S. P. Dajkhosh**, C. Preza, J. C. Barreiro-Hervas, G. Saavedra. *"Structured Illumination Microscopy by use of a Spatially Incoherent Source with*

Arbitrary Shape." **Focus on Microscopy**, 2022.

- *Poster*, **International Gordon Research Conference (GRC)**, Maine: "Performance Evaluation of 3D Physics-Informed Channel-Attention GAN on 3D SIM Datasets," Jun 2022.
- *Poster*, **EECE 8th Annual Poster Competition (2nd Place)**, Memphis, TN: "Physics-Guided Deep Learning Methodologies in Computational Imaging," Apr 2022.
- *Presenter*, **Institute for Intelligent Systems Python Workshop**, Memphis, TN: "Knowledge-Based Deep Learning Modeling," Oct 2021.
- *Contributor*, A. Zahedifar, S. P. Dajkhosh, M. Hosseinzadeh, Y. Ghanbari. **Neuromatch 3.0**: "Enhancing classification of autism spectrum disorder using resting-state fMRI with ML and dimensionality reduction," Oct 2020.

Selected Projects

- **Camera Calibration & Virtual Image Dimension Estimation**, Digital Image Processing: projection-matrix-based reconstruction in Python/OpenCV. (A, 2022)
- **Fake-Review Classification**, Machine Learning: neural networks, random forests, and logistic regression on Amazon product reviews in Python (Scikit-learn, Keras). (A, 2022)
- **Independent Study: DL Architectures for Microscopy Image Restoration**, review and re-implementation of multiple SOTA architectures and datasets in PyTorch/Keras. (A, 2022)
- **Physics-Guided DL for Image Restoration**: comprehensive review and SIM simulation toolkit in MATLAB. (A, 2021)
- **MIROSOT Robot, AUTCUP Senior League**, Parsian Robotics Lab, Amirkabir University of Technology, 2016: designed and implemented the full embedded hardware stack - FPGA processor board (Verilog/Xilinx), ARM Cortex-M4 motor controller, motor driver circuits, and inter-board communication. Competed at international AUTCUP Senior league.

Affiliations & Memberships

- **TBP (Tau Beta Pi)**, member, University of Memphis chapter, since Nov 2024.
- **V.I.S.A.**, member, Visitors and International Students Association, Memphis, since Nov 2023.
- **SPIE**, International Society for Optics and Photonics, member since Dec 2022.
- **IEEE-HKN (Eta Kappa Nu)**, member, University of Memphis chapter, since Dec 2022.

Languages

- **Persian**: native.
- **English**: full professional proficiency.
- **French**, **Arabic**: intermediate.